

# **Transcript Package User Guide**

**User Guide for Release 2026.08**

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## 1 TranscriptPkg and TranscriptBasePkg Overview

TranscriptPkg allows different parts of a testbench to use a common transcript file. As such, it provides means to open (TranscriptOpen) and close (TranscriptClose) the transcript file. In addition, it provides a variety of printing capabilities.

## 2 TranscriptPkg Use Models

All OSVVM printing uses the transcript capability. If the TranscriptFile is open, all OSVVM printing will go to it. If the TranscriptFile is not open all OSVVM printing will go OUTPUT (stdout). If the TranscriptFile is open and transcript mirroring is enabled, printing will go to both the TranscriptFile and to OUTPUT.

The following shows the basic use model. Set the test name (using SetTestName). Open the transcript file (using TranscriptOpen). Mirror the transcript file to OUTPUT (SetTranscriptMirror). Print using the transcribing capability. Close the transcript file (TranscriptClose) just before the end of the test. Generate reports (using EndOfTestReports). Stop the test case (using std.env.stop).

```
library osvvm ;
context osvvm.OsvvmContext ;
. . .

architecture Test1 of TestCtrl is
begin
  ControlProc : process
  begin
    SetTestName("Tb_SendGet_1") ;
    TranscriptOpen ;           -- Print to Tb_SendGet_1.log
    SetTranscriptMirror ;      -- Also print to OUTPUT
    PrintLine("Print 1") ;
    swrite(buf, "Write 1") ;
    writeline(buf) ;
    . . .
    TranscriptClose ;
    EndOfTestReports ;
    std.env.stop ;
  end process ControlProc ;
```

## 3 Method Reference

### 3.1 Package Reference

The entire OSVVM utility library, including TranscriptPkg can be referenced as follows:

```
library osvvm ;
context osvvm.OsvvmContext ;
```

### 3.2 Transcript File

TranscriptFile is declared in TranscriptBasePkg and is not visible to user programs.

```
file TranscriptFile : text ;
```

### 3.3 SetTestName

Formally this method is in AlertLogPkg. It gives the test case a name.

### 3.4 TranscriptOpen and TranscriptClose

TranscriptOpen opens the TranscriptFile. There are a number of variations (overloaded version) of TranscriptOpenWhen, however, the simplest is the following. It opens a file named TestCaseName.log, where TestCaseName is the name used in SetTestName.

```
procedure TranscriptOpen ;
```

TranscriptClose closes the TranscriptFile.

```
procedure TranscriptClose ;
```

When FILE\_OPEN\_KIND is not specified in the call, the first call to TranscriptOpen will open the file in WRITE\_MODE. If the transcript is closed (with TranscriptClose) and then reopened, it will be opened with APPEND\_MODE.

### 3.5 Additional overloading for TranscriptOpen

The following versions of TranscriptOpen use the TestCaseName.log as the file name.

```
procedure TranscriptOpen (OpenKind: WRITE_APPEND_OPEN_KIND) ;  
procedure TranscriptOpen (Status: InOut FILE_OPEN_STATUS) ;  
procedure TranscriptOpen (Status: InOut FILE_OPEN_STATUS;  
                          OpenKind: WRITE_APPEND_OPEN_KIND) ;
```

The following versions of TranscriptOpen allow the file name to be specified.

```
procedure TranscriptOpen (ExternalName: STRING) ;  
procedure TranscriptOpen (ExternalName: STRING; OpenKind: WRITE_APPEND_OPEN_KIND);  
procedure TranscriptOpen (Status: InOut FILE_OPEN_STATUS; ExternalName: STRING) ;  
procedure TranscriptOpen (Status: InOut FILE_OPEN_STATUS; ExternalName: STRING;  
                          OpenKind: WRITE_APPEND_OPEN_KIND) ;
```

### 3.6 SetTranscriptMirror

SetTranscriptMirror sets mirror mode in which case all OSVVM printing goes to both the TranscriptFile and std.textio.OUTPUT.

```
procedure SetTranscriptMirror (A : boolean := TRUE) ;  
...  
SetTranscriptMirror ;
```

### 3.7 IsTranscriptOpen

IsTranscriptOpen returns true when the TranscriptFile is open.

```
impure function IsTranscriptOpen return boolean ;
. . .
if IsTranscriptOpen then
    swrite(buf, "Write 1 - transcript open") ;
    writeline(TranscriptFile, buf) ;
end if;
```

### 3.8 IsTranscriptMirrored

IsTranscriptMirrored returns true when the transcript is mirrored.

```
impure function IsTranscriptMirrored return boolean ;
. . .
if IsTranscriptMirrored then
    . . .
```

### 3.9 WriteLine

Writeline writes the buffer using the transcript capability.

```
procedure WriteLine(buf : inout line) ;
. . .
swrite(buf, "A Message") ;
WriteLine(buf) ;
```

### 3.10 Print

Print writes a string using the transcript capability.

```
procedure Print(s : string) ;
. . .
Print("Another Message") ;
```

### 3.11 Blank

Blank puts a variable number of blank lines in the output.

```
procedure Blank (count : natural := 1) ;
. . .
BlankLine(5) ; -- prints 5 blank lines
```

### 3.12 PrintLine

Print prints the string using the transcript capability. Like Alert and Log, PrintLine prefixes the line with OSVVM\_PRINT\_PREFIX. In addition, like Log, print allows message filtering using Level and AlertLogID. Unlike Log, PrintLine does not print the time, "log", or the Log level.

```
procedure PrintLine (s : string) ;  
procedure PrintLine (s : string ; Level : LogType) ;  
procedure PrintLine (AlertLogID : AlertLogIDType ; s : string ; Level : LogType);
```

### 3.13 PrintHeader

Print prints the string plus prefix and suffix lines to frame the message and make it stand out. Each character in the prefix and suffix indicates a line to print. Each line printed is prefixed by OSVVM\_PRINT\_PREFIX. If the character is a space, then print a blank line, otherwise, repeat that character OSVVM\_LINE\_LENGTH times. In addition, like Log, PrintHeader supports message filtering using Level and AlertLogID.

```
procedure PrintHeader (s, Prefix, Suffix : string) ;  
procedure PrintHeader (s, Prefix, Suffix : string ; Level : LogType ) ;  
procedure PrintHeader (AlertLogID : AlertLogIDType ; s, Prefix, Suffix : string ;  
Level : LogType ) ;
```

There is also a short form that uses OSVVM\_HEADER\_PREFIX (default " ") and OSVVM\_HEADER\_SUFFIX (default "\*") instead of having a Prefix and Suffix parameter.

```
procedure PrintHeader (s : string) ;  
procedure PrintHeader (s : string ; Level : LogType ) ;  
procedure PrintHeader (AlertLogID : AlertLogIDType; s : string; Level : LogType);
```

### 3.14 HeaderLine

HeaderLine prints a string of characters that indicate a line to print. Each line printed is prefixed by OSVVM\_PRINT\_PREFIX. If the character is a space, then print a blank line, otherwise, repeat that character OSVVM\_LINE\_LENGTH times. In addition, like Log, HeaderLine supports message filtering using Level and AlertLogID.

```
procedure HeaderLine (S : string) ;  
procedure HeaderLine (S : string; Level : LogType; Enable : boolean := FALSE) ;  
procedure HeaderLine (AlertLogID : AlertLogIDType; S : string;  
Level : LogType := ALWAYS; Enable : boolean := FALSE) ;
```

### 3.15 BlankLine

BlankLine puts a variable number of blank lines in the output. Each line is prefixed by OSVVM\_PRINT\_PREFIX. Blank lines can also be filtered with Log levels and AlertLogID.

```
procedure BlankLine (count : natural := 1) ;  
procedure BlankLine (count : natural; Level : LogType; Enable : boolean := FALSE);  
procedure BlankLine (AlertLogID : AlertLogIDType ; count : natural;  
Level : LogType := ALWAYS; Enable : boolean := FALSE);
```

## 4 About TranscriptPkg

TranscriptPkg is part of the OSVVM Utility library.

TranscriptPkg was created by Jim Lewis of SynthWorks. Please support our work by buying your VHDL Training from SynthWorks.

TranscriptPkg.vhd is a work in progress and will be updated from time to time. Caution, undocumented items are experimental and may be removed in a future version.

## 5 Compiling and Using OSVVM Utility Library

Compilation order for OSVVM Utility Library is in CHANGELOG.md and osvvm.pro. To run osvvm.pro see the Script User Guide.

OSVVM Utility library is released under the Apache open source license. It is free (both to download and use - there are no license fees). You can download it from osvvm.org or from our development area on GitHub (<https://github.com/OSVVM/OSVVM>).

If you add features to the package, please donate them back under the same license as candidates to be added to the standard version of the package. If you need features, be sure to contact us.

We also support the OSVVM user community and blogs through <http://www.osvvm.org>. Interested in sharing about your experience using OSVVM? Let us know, you can blog about it at osvvm.org.

## 6 About the Author - Jim Lewis

Jim Lewis, the founder of SynthWorks, has thirty plus years of design, teaching, and problem solving experience. In addition to working as a Principal Trainer for SynthWorks, Mr Lewis has done ASIC and FPGA design, custom model development, and consulting.

Mr. Lewis is the primary developer of the Open Source VHDL Verification Methodology (OSVVM.org) packages and a participant in the IEEE VHDL standards. Neither of these activities generate revenue.

Please support our volunteer efforts by buying your VHDL training from SynthWorks.

If you find bugs these packages or would like to request enhancements, you can reach me at [jim@synthworks.com](mailto:jim@synthworks.com).